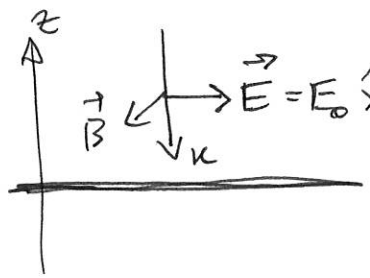
Relate optical conductivity $\sigma = \frac{e^2}{4\hbar}$
to simple observables:

<u>EM wave normally incident on graphene layer</u>



graphene layer at $z=0$

Maxwell eqns $\begin{cases} \vec{\nabla} \times \vec{E} = -\frac{1}{c}\frac{\partial \vec{B}}{\partial t} \\ \vec{\nabla} \times \vec{B} = \frac{4\pi}{c}\vec{j} + \frac{1}{c}\frac{\partial \vec{E}}{\partial t} \end{cases}$

$$\underbrace{\vec{\nabla} \times \vec{\nabla} \times \vec{E}}_{\vec{\nabla}(\vec{\nabla}\cdot\vec{E}) - \vec{\nabla}^2 \vec{E}} = -\frac{1}{c}\frac{\partial}{\partial t}\vec{\nabla}\times\vec{B} = -\frac{4\pi}{c^2}\frac{\partial \vec{j}}{\partial t} - \frac{1}{c^2}\frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\frac{1}{c^2}\frac{\partial^2 \vec{E}}{\partial t^2} - \vec{\nabla}^2 \vec{E} = -\frac{4\pi}{c^2}\frac{\partial \vec{j}}{\partial t} = -\frac{4\pi\sigma}{c^2}\frac{\partial \vec{E}}{\partial t}\delta(z)$$

$$\vec{j} = \sigma \vec{E}\,\delta(z)$$ (current only flows in the layer at $z=0$)

Incoming $\vec{E}_I = E_0 \hat{x} \cos(\omega t + \frac{\omega z}{c})$

Reflected $\vec{E}_R = E_R \hat{x} \cos(\omega t - \frac{\omega z}{c})$

Transmitted $\vec{E}_T = E_T \hat{x} \cos(\omega t + \frac{\omega z}{c})$

<u>BC</u> $E_I + E_R = E_T$ (1) (tangential components)

$$\frac{\omega^2}{c^2}E + \frac{d^2 E}{dz^2} = \frac{4\pi\sigma}{c^2}\left(-E\,\omega \sin(\omega t + \frac{\omega z}{c})\right)\delta(z)$$

$$= -\frac{4\pi\sigma\omega}{c^2} E \sin(\omega t)\,\delta(z).$$

Integrate both sides of the equation ($\delta \to +0$)

$$\int_{-\delta}^{\delta} dz (\quad) = -\frac{4\pi\sigma\omega}{c^2}\int_{-\delta}^{\delta} dz\, E \sin(\omega t)\,\delta(z)$$

$$\Rightarrow \quad E'(\delta) - E'(-\delta) = -\frac{4\pi\omega\sigma}{c^2} E_T \sin(\omega t)$$

$$E'(\delta) = \frac{d}{dz}\left(E_0 \hat{x} \cos\left(\omega t + \frac{\omega z}{c}\right) + E_R \hat{x} \cos\left(\omega t - \frac{\omega z}{c}\right)\right)_{z=0} =$$

$$= -E_0 \hat{x} \frac{\omega}{c} \sin(\omega t) + E_R \hat{x} \frac{\omega}{c} \sin(\omega t).$$

$$E'(-\delta) = \frac{d}{dz}\left(E_T \hat{x} \cos\left(\omega t + \frac{\omega z}{c}\right)\right)_{z=0} = -E_T \hat{x} \frac{\omega}{c} \sin(\omega t).$$

$$-E_0 \frac{\omega}{c} \sin(\omega t) + E_R \frac{\omega}{c} \sin(\omega t) - \left(-E_T \frac{\omega}{c} \sin(\omega t)\right) =$$

$$= -\frac{4\pi\omega\sigma}{c^2} E_T \sin(\omega t).$$

$$-E_0 + E_R + E_T = -\frac{4\pi\sigma}{c} E_T \quad (2)$$

$$\begin{cases} E_0 - E_R = \left(1 + \frac{4\pi\sigma}{c}\right) E_T \\ E_0 + E_R = E_T \end{cases}$$

$$2E_0 = \left(2 + \frac{4\pi\sigma}{c}\right) E_T$$

$$\frac{E_0}{E_T} = 1 + \frac{2\pi\sigma}{c}$$

Transmittion coefficient $T = \left|\frac{E_T}{E_0}\right|^2 = \frac{1}{\left(1 + \frac{2\pi\sigma}{c}\right)^2}$

$$T \approx \frac{1}{\left(1 + \frac{2\pi e^2}{4\hbar c}\right)^2} \approx 1 - \pi\left(\frac{e^2}{\hbar c}\right) = 0.977$$

$\rightarrow$ fine structure constant

To consider effect of uniform magnetic field

$$\vec{A} = \frac{1}{2} B(-y, x, 0) = \frac{1}{2} B(-y\hat{x} + x\hat{y})$$

$$\vec{\nabla} \times \vec{A} = B\hat{z}.$$

Then $\vec{p} \to \vec{p} - \frac{e}{c}\vec{A} = \begin{cases} \hat{p}_x - \frac{e}{c}\left(-\frac{By}{2}\right) \\ \vec{p}_y - \frac{e}{c}\left(\frac{Bx}{2}\right) \end{cases}$

But $[\hat{p}_x, \hat{p}_y] \Rightarrow [\hat{p}_x + \frac{eBy}{2c}, \hat{p}_y - \frac{eBx}{2c}] =$

$$= -\frac{eB}{2c}\underbrace{[\hat{p}_x, x]}_{-i\hbar} + \frac{eB}{2c}\underbrace{[y, \hat{p}_y]}_{i\hbar} = i\frac{eB\hbar}{c} = -i\frac{|e|\hbar B}{c}$$

let now $\hat{p}_x - i\hat{p}_y = \sqrt{\frac{2|e|\hbar B}{c}}\,\hat{a}$

$$\hat{p}_x + i\hat{p}_y = \sqrt{\frac{2|e|\hbar B}{c}}\,\hat{a}^+$$

~~then $[p_x - ip_y, p_x + ip_y] = i\frac{eB\hbar}{c}$~~

$$[\hat{p}_x - i\hat{p}_y, \hat{p}_x + i\hat{p}_y] = i[p_x, p_y] - i[p_y, p_x] = \frac{2|e|\hbar}{c}B$$

$$\| $$

$$\frac{2e\hbar B}{c}[\hat{a}, \hat{a}^+] = \frac{2e\hbar B}{c}$$

$$\Rightarrow \hat{H}_B = v\vec{\sigma}\cdot\vec{p} = v\begin{pmatrix} 0 & p_x - ip_y \\ p_x + ip_y & 0 \end{pmatrix} = v\sqrt{\frac{2e\hbar B}{c}}\begin{pmatrix} 0 & \hat{a} \\ \hat{a}^+ & 0 \end{pmatrix}$$

let $l_B = \sqrt{\frac{\hbar c}{eB}}$ magnetic length $\Rightarrow \frac{\sqrt{2}\hbar v}{l_B}\begin{pmatrix} 0 & \hat{a} \\ \hat{a}^+ & 0 \end{pmatrix}$

$$\hat{H}_B \Psi = \varepsilon \Psi \quad , \Psi = \begin{pmatrix} u \\ v \end{pmatrix}$$

$$\begin{pmatrix} 0 & \hat{a} \\ \hat{a}^+ & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \varepsilon' \begin{pmatrix} u \\ v \end{pmatrix} \quad , \varepsilon' \equiv \frac{\varepsilon}{v\sqrt{\frac{2e\hbar B}{c}}}$$

$$\begin{cases} \hat{a} v = \varepsilon' u \\ \hat{a}^+ u = \varepsilon' v \end{cases}$$

$$\Rightarrow \hat{a}^+ \left(\frac{\hat{a} v}{\varepsilon'} \right) = \varepsilon' v \Rightarrow \hat{a}^+ \hat{a} v = (\varepsilon')^2 v$$

$\Rightarrow$ second spinor component (upto a normalization) is an eigenstate of number operator $v = v_n = |n\rangle$

$$\Rightarrow (\varepsilon'_n)^2 = n$$

$$\varepsilon'_n = \pm \sqrt{n}$$

$$\frac{\varepsilon_n}{v\sqrt{\frac{2e\hbar B}{c}}} = \pm \sqrt{n} \quad , \varepsilon_n = \pm v \sqrt{\frac{e\hbar B}{c}} \sqrt{2n}$$

$$= \pm \hbar v \sqrt{\frac{eB}{\hbar c}} \sqrt{2n}$$

$$= \pm \frac{\hbar v \sqrt{2}}{\ell_B} \sqrt{n}$$

$$\varepsilon_{\lambda, n} = \lambda \hbar \omega_c \sqrt{n}$$

$\lambda = \pm 1$, $\hbar \omega_c = \hbar v \sqrt{2} / \ell_B$ = cyclotron freq.

Notice: $\varepsilon' u_n = \hat{a} v_n = \sqrt{n}\,|n-1\rangle$!

since $\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle$

But this assumes that state $|n-1\rangle$ exists!

$\Rightarrow$ correct only for $n \geq 1$.

Thus for $\underline{n=0}$ (zeroth LL)

$$v_n = |n\rangle \Rightarrow v_0 = |0\rangle$$

$$\text{but } u_n = \frac{1}{\varepsilon'}\hat{a}\,|0\rangle = 0$$

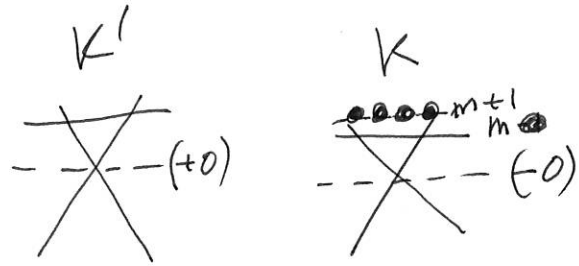
$$\psi_{n=0} = \begin{pmatrix} 0 \\ |n=0\rangle \end{pmatrix} \quad \text{but} \quad \psi = \begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix}$$

$\Rightarrow$ in this case all electrons are on the B-sublattice!

normalized $\psi_{n>0} = \frac{1}{\sqrt{2}}\begin{pmatrix} |n-1\rangle \\ |n\rangle \end{pmatrix}$

Because $\varepsilon' = \sqrt{n} \Rightarrow u_n = \frac{1}{\varepsilon'}\hat{a}\,|n\rangle = \frac{\sqrt{n}}{\sqrt{n}}\,|n-1\rangle = |n-1\rangle$

and $\frac{1}{\sqrt{2}}$ follows from normalization.



Completely filled upto level m

$$\frac{E_K}{D} = \sum_{n=\infty}^{0} -\hbar\omega_B\sqrt{n} + \sum_{n=1}^{m} \hbar\omega_B\sqrt{n} + \text{partially filled}$$

$$\frac{E_{K'}}{D} = \sum_{n=\infty}^{1} -\hbar\omega_B\sqrt{n} + \sum_{n=0}^{m} \hbar\omega_B\sqrt{n} + \text{part. filled}$$

Since degeneracy $D = 4A\frac{eB}{2\pi\hbar c} \sim B$, and $\omega_B = \frac{\sqrt{2}v}{l_B} \sim \sqrt{B}$

we observe that $E \sim (B)^{3/2}$

$\Rightarrow$ magnetization $M = -\frac{\partial E}{\partial B} \sim B^{1/2}$

susceptibility $\chi = \frac{\partial M}{\partial B} \sim B^{-1/2} \underset{B\to 0}{\to} \infty$!

Physical picture for neutral graphene ($\mu=0$):

$N_0 = D\cdot$ states around zero energy

coalese into $n=0$ Landau level

$$N_0 = \int_{-\lambda}^{\lambda} d\varepsilon\, \nu(\varepsilon) = \frac{4A}{2\pi\hbar^2 v^2}\lambda^2 = \frac{4A\,eB}{2\pi\hbar c}$$

$\hookrightarrow 4A\frac{|\varepsilon|}{2\pi\hbar^2 v^2}$ (DOS)

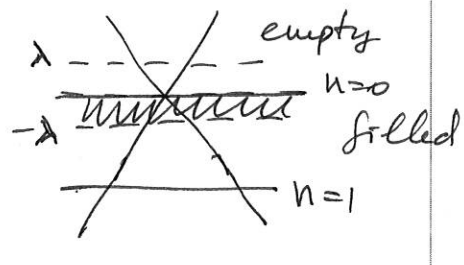
$$\Rightarrow \lambda = \hbar v\sqrt{\frac{eB}{\hbar c}} = \frac{\hbar v}{l_B} = \frac{\hbar\omega_B}{\sqrt{2}}$$

Energy of these states without the magnetic field

$$E_0 = \int_{-\lambda}^{0} d\varepsilon\, \varepsilon\, \nu(\varepsilon)$$

$$E_0 = -\int_{-\lambda}^{0} d\varepsilon \; \varepsilon \; \frac{4A(-\varepsilon)}{2\pi\hbar^2 v^2} = \frac{4A}{2\pi\hbar^2 v^2} \; \frac{1}{3}\varepsilon^3 \Big|_{-\lambda}^{0} =$$

$$= -\frac{4A}{6\pi\hbar^2 v^2}\left(\frac{\hbar\omega_B}{\sqrt{2}}\right)^3 = -\gamma B^{3/2}$$



When mag. field is on,
all these N_0 states are at 0 (zero) energy

$\Rightarrow \Delta E = 0 - E_0 = \gamma B^{3/2}$ (i.e. energy cost is positive!)

$$M = -\frac{\partial \Delta E}{\partial B} = -\frac{3}{2}\gamma B^{1/2}$$

$$\chi_{orb} = \frac{\partial M}{\partial B} = -\frac{3\gamma}{4}\frac{1}{B^{1/2}} \xrightarrow[B\to 0]{} \infty$$

McClure result (1956): diverging linear susceptibility

$$\chi_{orb} = -\left(\frac{g\mu_B}{2}\right)^2 \frac{2\cdot 4\cdot m_e^2 v^2}{3\pi\hbar^2}\,\delta(\varepsilon_F)$$

delta-function!

$$\mu_B = \frac{e\hbar}{2m_e c}$$ $\Rightarrow$ no dependence on m_e mass!